

Unit 08: Network layer - logical addressing

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Objectives

- what is an IP address?
- types of IP address.
- how to find the public IP address
- classification of IP addresses
- comparison between ipv4 and ipv6
- understand Classful Addressing and classless addressing
- understand NAT, ICMP, ARP, and RARP

Introduction

How can you locate your computer's or smartphone's IP address? Computers on the internet connection with one another through underground or underwater cables, as well as wirelessly. If I want to copy a file from the internet, I think my machine has to have a URL so that other computers on the internet can identify and locate it. In Internet terms, a computer's address is known as an IP address; an IP address is essentially a device's identity; without an IP address, no system can connect with another on the internet. Let's look at another case to better explain what an IP address is. Assume if anyone wishes to deliver mail here and needs a home address. You must have a home address to deliver mail to others. Similarly, whether you want to send or communicate with the machine's programs, or if you want to copy something from any internet-connected device to your machine, the computer requires an address as seen in Figure 1.

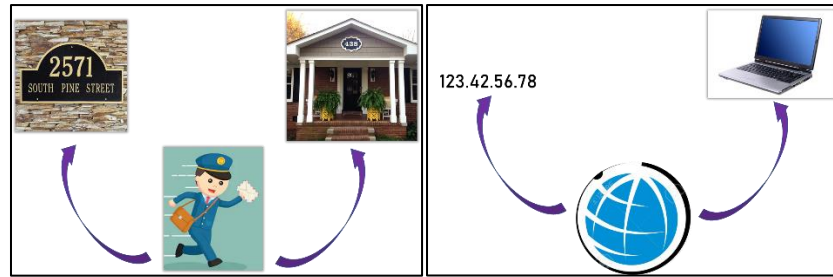


Figure 1 Comparison of a home address with IP address

Such that machines on the internet give you a file that you wish to download, that address is called an IP address in internet terms. An IP address is nothing more than a set of numbers written in a certain format. Now, this is your home address, as determined by your local postman, and it is to this address that you receive all mail. And your home's identity is based on its address. Someone would like to deliver a message. So, the postman will send the letter to you because they know this is the address and we have to bring this letter to this same address. Similarly, the machine has an IP address that is used to identify it on the internet, and you have access to all downloadable files. This computer's address is: As a result, this is the system's IP address.

8.1 IP address and its full form

The Internet Protocol Address is the entire version of the IP address. What exactly is the concept of Internet Protocol? The Internet Protocol (IP) is a series of laws that govern how the Internet functions.

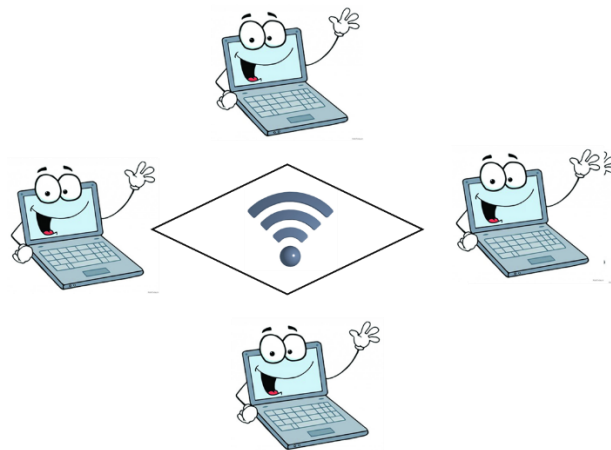


Figure 2 IP Address

Two Types of IP Addresses

So now we have two kinds of IP addresses, one for IP version four and the other for IP version six as seen in Figure 3.

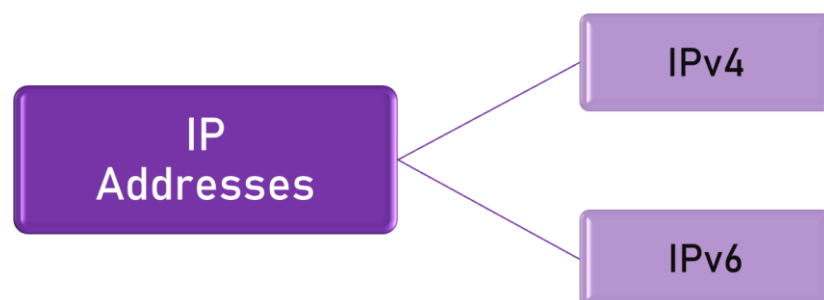


Figure 3 Two types of IP Address

What exactly is IP version 4? Why have we gone on to IP version six while we have IP version four? But, first and foremost, let's define IP version four. The full form of IP version four is internet protocol

version four, which consists of four numbers separated by a dot and dot n dot n dot n, which means there are four numbers separated by a dot operator as seen below.

N . N . N . N

0-255 . 0-255 . 0-255 . 0-255

The IP address is made up of numbers in the range of zero 0 to 255. On 192.168.0.1, I can write. However, if I send a message to 256.168.0.1, this is an invalid IP address since the range should be between 0 and 255. However, machines are incapable of comprehending decimal numbers. Since we're written in decimal numbers, then every range we're writing in between 0 and 255 is in decimal numbers. Computers, on the other hand, do not grasp decimal numbers so they are perplexed by them. So, what is the answer to this? As a result, we must translate these decimal numbers into binary numbers that the machine can comprehend. As a result, the binary spectrum can be written as shown in Figure 4.

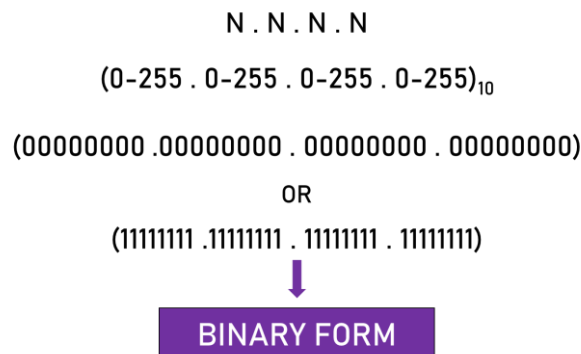


Figure 4 IP in Binary Form

So IP version four is a 32-bit address, which is a special sequence of ones or zeros allocated to each computer; we call it unique since no two devices will use the same IP address as seen in Figure 5.

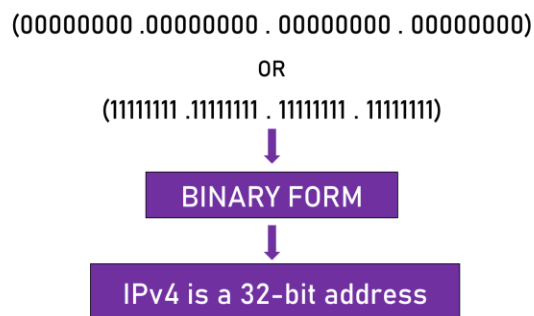


Figure 5 IP is a 32-bit address

That is why it is referred to as a specific address. Any device connecting to the internet will have a different IP address. They have a specific IP address any time they connect to the internet. So, the question now is how many devices they can address. As a result, a total of $2^{\text{power}32}$, or roughly 4 billion computers, can be addressed and wired to the internet. Version four of IP. But it's only capable of addressing 4 billion devices. As a result, we can only bind 4 billion computers as shown below.

$$2^{32} = 4,294,967,296 \text{ Devices}$$

ISSUE with 32-bit IPv4

Nowadays, every machine, every user, has a laptop, tabs, and several devices; they have a desktop, their desktop, and they use smartphones as well. As a result, they like the IP address of any device. Also in today's world, not every machine is present in every household. Everyone has their scheme. Everyone has their smart screen much of the time. They want to connect to the internet, and to do so, they would need IP addresses. As a result, IP version four will not be able to supply IP addresses to all of them because it can only communicate with 4 billion addresses. As a result, it can only have 4 billion addresses. So far, we've surpassed the 4 billion mark, and we're on our way to IP version six.

IPv6

Let's take a look at IP version six now. IP version 6 is a 128-bit address written as a series of eight hexadecimal digits separated by the column in human-readable mode. IP version six can be written as, as you can see here, in a human-readable format. However, computers do not understand hexadecimal, so we must convert it back to binary. It'll be divided into 128 ones and zeros. Since each device linked to the internet is given this sequence of zeros and ones, since no two machines should be allocated the same IP address, this is a special series number. In this example, IP version four is the same, so both would be unique. With IP version six, 2^{128} computers to the internet.

This is the value that has been seen here: 2 power of 128 equals as seen in Equation 1. So, with this many addresses, it will address this many devices. This importance indicates that it can handle this many computers, which is much more than sufficient for future generations.

$$2^{128} = 340,282,366,920,938,463,463,374,607,431,768,211,456$$

Equation 1

How to Find Your Computer Public IP Address?

So, first and foremost, go to Google and type in "what is my IP." As you can see, we've opened Google and are typing in "what is my IP." As a result, it will show your public IP address. Even now, the only concern is how you'll search your smartphone's public IP address.

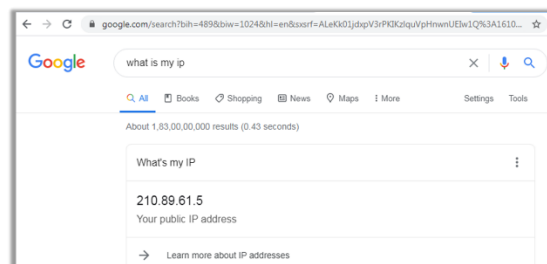


Figure 6 What is my IP

So, in Google, write down my IP address. As a result, Google will inform you of your smartphone's public IP address. One very important thing to remember is that we won't be able to connect to the internet without IP versions four and six; an IP address is needed for an internet connection. You believe that if you don't have an IP address, you won't be able to log in to the internet, which means you won't be able to download or upload something, and you won't be able to check something. But you'll need an IP address if you want to connect to the internet.

Classification of IP Addresses

IP addresses are divided into two categories: dynamic IP addresses and static IP addresses.

Dynamic IP Address

When you bind your device or smartphone to the internet, this is what happens. Then there's Internet Service Provider ISP, which is a company that provides an IP address from a pool of valid IP addresses. You have an IP address, then you can connect to the internet. Once connected, you can transmit and receive data to and from other devices on the internet. So, the next time you want to connect to the Internet, do so. ISP will assign you a new IP address with the same availability set, which is why it is called dynamic. Dynamic means that it changes all of the time. DHCP assigns a unique IP address to every device.



Figure 7 Dynamic IP Address

Static IP Address

It is the only other classification that remains constant. Who is using this static IP address and who are the domain name servers that are using this IP address? What is the DNS server actually, these are the computers that help you to open a website on your computer?

A static IP address provides information, such as the device is located in which continent. In which country, which city, and which is the internet service provider, that is providing the internet connection to that device.



Figure 8 Example of Static IP Address

It is an internet service provider that assigns IP addresses to computers so that they can connect to the Internet. We can monitor the location of a computer connecting to the internet until we know the ISP. As a result, IP addresses allow billions of devices to be identified. What is the disadvantage of using a static IP address? One of the most significant disadvantages is that it is less safe so it is easy to trace. As a result, we can classify it as less stable.

8.2 Comparison between IPv4 and IPV6

Table 1 shows the differences between IPv4 and IPv6.

Table 1 Comparison between IPv4 and IPv6

IPv4	IPv6
IPv4 uses a 32-bit address for its Internet addresses	IPv6 utilizes 128-bit Internet addresses.
it can provide support for 2^{32} IP addresses	it can support 2^{128} Internet addresses
IPv4 is a numeric address, and its binary bits are separated by a dot (.)	IPv6 is an alphanumeric address whose binary bits are separated by a colon (:). It also contains hexadecimal.
Number of header fields 12	Number of header fields 8
Has checksum fields	Does not have checksum fields
Types of addresses are Unicast, broadcast, and multicast.	Unicast, multicast, and anycast.
Networks need to be configured either manually or with DHCP. IPv4 had several overlays to handle Internet growth, which require more maintenance efforts.	IPv6 support autoconfiguration capabilities.

8.3 Classful Addressing

From 1981 before Classless Inter-Domain Routing was implemented in 1993, classful addressing was a network addressing the Internet's architecture. Centered on four address bits, this addressing system splits the IP address into five distinct groups. Group A, B, C, D, and E are the five categories in which classful addressing is divided. An IP address is a 32-bit unique address with a 2^{32} address space.

In general, IP addresses are written in one of two ways: dotted decimal notation or hexadecimal notation as seen in Figure 9. It means classful addressing can be represented in binary also, and it can be represented as a decimal also.

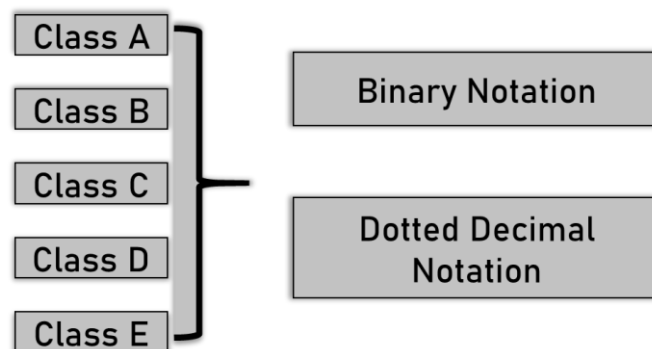


Figure 9 Classful Addressing

Binary Notation

In binary notation, the IP version four address is displayed as 32-bits each octet is often referred to as a byte. So, it is common to hear an IP version four address refers to as a 32-bit address or a four-byte address.

The following is an example of an IPv4 address in binary notation:

01110101 10010101 00011101 00000010

As a result, the IP version four address has been compressed and made easier to read. Internet addresses are normally written with the decimal point dividing the bytes in decimal form.

Decimal Notation

So, we have two tools, one of which is binary and the other of which is decimal, and the same binary is translated into decimal notation. So, there are two types of notations: dotted decimal and binary.

Dotted-decimal notation and binary notation for an IPv4 address



Figure 10 Notation for IPv4

As a result, the IP version four address has been compressed and made easier to read. Internet addresses are normally written with the decimal point dividing the bytes in decimal form as seen in Figure 10.

Important points to remember

1. There must be no leading zero (045).
2. There can be no more than four numbers in an IPv4 address.
3. Each number needs to be less than or equal to 255 (301 is outside this range).

255.272.255.255

4. A mixture of binary notation and dotted-decimal notation is not allowed

172. 00101000. 254. 11001100

Find the error, if any, in the following IPv4 addresses

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

- When writing in decimal notation, it cannot be begun from zero like it is in the first one.
- It is divided into five octets in the B, so five octets are not allowed.
- The range for the third one should be between 0 and 255. The 301 series is not accurate.
- The fourth one uses a combination of binary and dotted decimal notation.

How would you be able to know to which class it belongs

When your IP address is written in binary notation and the first bit is 0, it means it's a class A address. If the first two bits of your IP address are one zero, your IP address is classified as Class B. If the first three bits of the binary notation are 110, It is classified as Class C. It belongs to Class D if the first four bits are 1111 as seen in Figure 11.

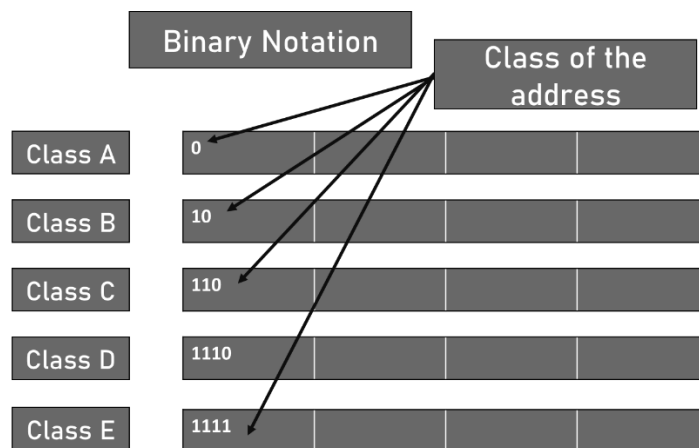


Figure 11 Class of the address in Classful Addressing

Example of Binary Notation

So, since the first bit is zero, it is classified as Class A. Let's look at another case. So far, only the first three bits have been discovered. The first two bits are one, and the third bit is zero, indicating that this is a Class C address as seen below in Figure 12.

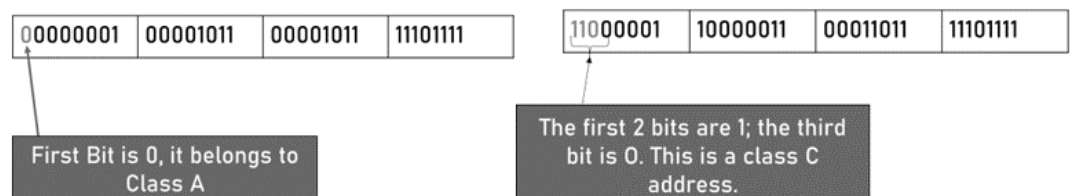


Figure 12 Example of Binary Notation

Example of Dotted Decimal Notation

If my IP address is expressed in decimal form. So, how can I determine the class it belongs to? So, there's a range, and it's divided into classes, and your IP addresses are divided into those classes. The size of Class A is 0 to 127. Class B has a range of 128 to 191, Class C has a range of 192 to 223 and Class D has a range of 224 to 239, and Class E has a range of 240 to 255.

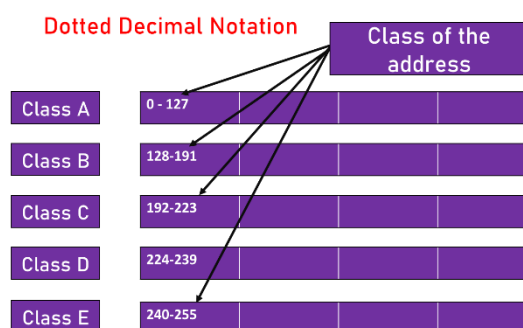


Figure 13 Dotted Decimal Notation

So, if the first byte is 14, we now have a total of 14 bytes. It indicates that they fall within the range of 0 to 127, which is class A. 252, and that they fall within the range of 240 to 255, which is class E.

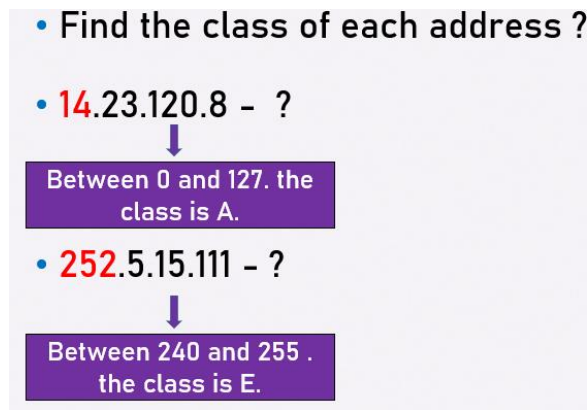


Figure 14 Example of decimal notation

Division of Classes

How can we know which part is a network ID and which part is a host ID if groups are split into two parts? The bits used for network ID and host ID, as well as the total number of networks and hosts possible in that class, are determined by the IP address class. Each computer connecting to a network is assigned an IP address by the ISP or network administrator. As a result, an IP address in class A, B, or C is split into Net ID and host ID in a classful addressing scheme. The length of these pieces varies depending on the address's class. So keep in mind that Class D and E are reserved addresses.

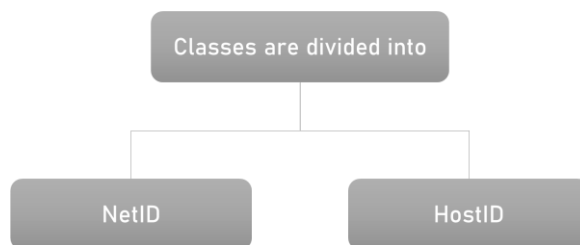


Figure 15 Division of Classes

Division in Class A

So, in class A, the first eight bits represent the net ID and the remaining bits represent the host ID; the remaining bits representing host ID, so the host ID is represented by 24 bits as seen in Figure 16.

NetID	HostID	HostID	HostID
0	7	8	31
0.0.0.0	to	0.255.255.255	
127.0.0.0	to	127.255.255.255	

We never assigned these to any hosts...these representing default network.

0 – default network

127---Loopback address...check whether NIC is working properly or not

Figure 16 Division in Class A

Important Points about Class A

- IP address begins with 0

- 7 remaining bits in the network part
 - Only 128 possible class A networks.
- 24 bits in the local part
 - Over 16 million hosts per class A network.
- All class A network parts are assigned or reserved

As you can see in Figure 16, zero to seven show the network element, so how many bits are there, total eight bits, and how many bits are there for the host, 24 bits, so each block has eight bits and each byte has eight bits, so eight plus eight plus eight equals 24 bits. But, as you can see, we're going to represent it in binary notation, as you can see here.

Exceptions in the class A IP addresses

The term "exception" refers to the fact that there are certain networks that we are unable to access, such as network 0, which is the default network. 127 is the loopback address, which is used to determine if the network interface card is functioning correctly or not. It is also known as the loopback address. Table 2 shows a summary of Class A IP Addresses.

Table 2 Summary of Class A IP Address

Class	Leading Bits	Size of network number field	Size of rest bit field	Number of networks	Address per network
Class A	0	8	24	128(2pow 7)	16,777,216 (2pow24)

Range Starts from

0.0.0.0 to 127.255.255.255

But in actual we can use
1.0.0.0 to 126.255.255.255

When a leading bit is 0, it indicates that the first bit should also be zero. The network number area is eight characters long. The remainder of the bit is 24 network area, which means that it is first in the network field, so all of the other bits represent the host, so 24 plus eight equals 32 number of networks 128 address per network two raised to the power of 24. But now we're using one to 126, which is the range of classes zero to 127, except zero, is reserved for the default network and 127 is the loopback address. But we might assume that we're using a scale of one to 126.

Division in Class B

The network portion is represented by the first 16 bits. The host part is represented by the next 16 pieces as seen in Figure 17. A balance between the number of networks and the number of hosts is achieved by using 16 bits for the network portion and 16 bits for the host part.

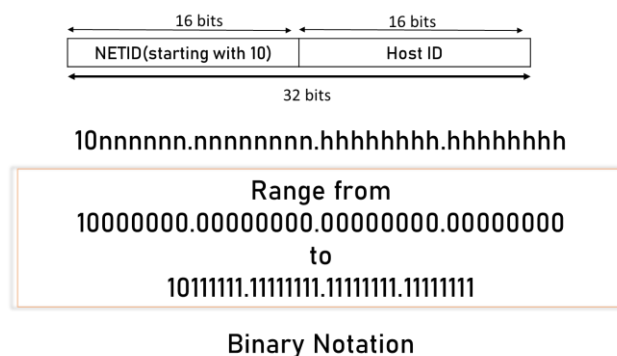


Figure 17 Division in Class B

Important Points in Class B

- 16 bits for network part 16 bits for the host part

- A balance between the number of networks and hosts.
- IP address begins with 10(1st zero in the second position)
- 14 remaining bits in the network part
 - Over 16,000 possible class B network
- 16 bits in the local part
 - Over 65,000 possible hosts

The network portion is represented by the first 16 bits. The host part is represented by the next 16 pieces. A balance between the number of networks and the number of hosts is achieved by using 16 bits for the network portion and 16 bits for the host part. So, in decimal notation, we'll write down more, such as several zero to 127 for class A, and 128 to 191 for class B. We've translated the binary to decimal, so 128 to 191 is the result as seen in Figure 18.

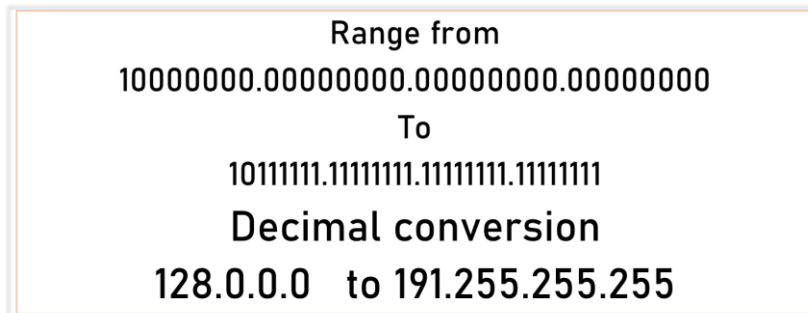


Figure 18 Conversion to Decimal Notation

Exceptions in Class B Addresses

- Exceptions in Class B IP Address
 - 169.254.x.x
- We never assigned this IP address to any host because this range reserve for APIPA.
 - 169.254.x.x is a private IP Addressing space reserved by Microsoft which it assigns automatically to your Network Adapter if it cannot get an IP Address from the DHCP Server

Table 3 has a list of Class B. The first two bits should be one zero, as we remember. Only then may we assume it is a Class B address, with a network number bit field of 16 bits and a host bit field of 16 bits. The number of networks can be represented by 2 power 14 and the number of addresses per network by 2 power 16. As a result, the scale is 128 to 191.

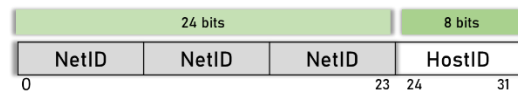
Table 3 Summary of Class B IP Address

Class	Leading Bits	Size of network number bit field	Size of text bit field	Number of networks	Address per network
B	10	16	16	16384(2 pow 14)	65536(2 pow 16)

128.0.0.0 to 191.255.255.255

Division in Class C

When it comes to class C, the first 24 bits represent the network, while the last eight bits represent the host as seen in Figure 19. It is a common and widely used IP address that starts with the 110 or the first zero in the third position in the case of a Class C address. There are over 2 million potential class C networks with 21 more bits in the network part and eight bits in the local part.



110nnnnn.nnnnnnnnn.nnnnnnnnn.hhhhhhhh

Range from

11000000.00000000.00000000.00000000

To

11011111.11111111.11111111.11111111

Binary Notation

Figure 19 Division in Class C

So, as you can see in the binary notation, the first three bits are set, which is 110, and the next eight bits are representing the network element, and the last eight bits are representing the host part, which we represented with the 111 and the 000, as in the binary notation here.

Important points in Class C

- 24 bits for the network address and 8 bits for host
- Popular and commonly used
- IP address begins with 110(1st zero in 3d position)
- 21 more bits in the network part
 - Over 2 million possible class C networks
- 8 bits in the local part
 - Only 256 possible hosts per class C network.

When we translate this to decimal notation, it becomes 192 to 223 as seen in **Error! Reference source not found.**

110nnnnn.nnnnnnnnn.nnnnnnnnn.hhhhhhhh

Range from

192.0.0.0 to 223.255.255.255

Decimal Notation

So, let's look at this overview of a Class C leading bits: 110 since it begins with 110, the size of the network number bit field is 24, the text bit field is eight, and the number of networks is two lift to power 21 because 110 is reserved out of 24 network bits. The number of addresses per network has been increased from 2 power 8 which equals to 256. The summary of Class C is shown in Table4.**Error! Reference source not found.**

Class D

The IP address range now begins with 224 and ends with 239, which is the range of IP addresses. The first four bits are always 1110, which is used for multicasting IP addresses and always starts with zero. As a result, it delivers a message to a network of hosts rather than just one. There are no network or host bits specified. The first four bits are always triple one, zero, as you can see. When we are expressed in binary form as seen in Table 4.

Table 4 Range of Class D

1110	Multicast(28 bit)
------	-------------------

Important Points in Class D

- Starts from 224.0.0.0 to 239.255.255.255
- There is no subnet mask for Class D.
- Used for multicasting (one to many types)
- IP address begins with 1110
- Used for multicasting, not defining networks
 - Sending a message to a group of hosts
 - Not just to one
 - Do not define network or host bits
 - Different from broadcasting
 - Used for videoconferencing.

As a result, when we translate this to decimal notation. As a result, the ranges are 224 to 239 as shown in Figure 20.

In binary

11100000.00000000.00000000.00000000

To

11101111.11111111.11111111.11111111

Decimal range

224.0.0.0 to 239.255.255.255

Figure 20 Decimal Conversion in Class D

Popular Class D IP Addresses

- 224.0.0.0: Base address reserved
- 224.0.0.1: Used for all multicasting host groups
- 224.0.0.2: Used for all subnet routers
- 224.0.0.5 and 224.0.0.6: Used by Open Shortest Path First, an interior gateway protocol for all network segment routing information

Class E

Class E IP addresses are only used for experimental and testing purposes. Class E IP addresses range from 240.0.0.0 to 255.255.255.254. There is no sub-net mask for this class. The first octet of class E's higher-order bits is always set to 11110 as shown in Table 5.

Table 5 Range of Class E

11110	Future use (27 bit)
-------	---------------------

Important points in Class E

- Class E IP addresses starts from 240.0.0.0. to 255.255.255.255
- Reserved for experimental purposes.
- IP address begins with 1111.

In Binary Form

11110000.00000000.00000000.00000000

To

11111111.11111111.11111111. 11111111

In decimal form

240.0.0.0

To

255.255.255.255

Range of Special IP addresses

169.254.0.0 – 169.254.0.16	Link-local addresses
127.0.0.0 – 127.0.0.8	Loop-back addresses
0.0.0.0 – 0.0.0.8	used to communicate within the current network.

Issues with Classful Addressing

There are many issues with classful addressing, the most serious of which is the lack of a network class capable of supporting a medium-sized domain efficiently. A Class C network with 254 hosts is typically too thin, whereas a Class B network with 65,534 hosts is much too big. A request for a network address block from a medium domain is typically addressed by assigning a Class B network address rather than several Class C network addresses, losing thousands of possible host addresses, and causing a detrimental effect on the Internet routing tables.

Design of Classful Addressing

- Class A addresses were designed for large organizations with a large number of attached hosts or routers.
- Class B addresses were designed for midsize organizations with tens of thousands of attached hosts or routers.
- Class C addresses were designed for small organizations with a small number of attached hosts or routers

8.4 Classless Addressing

Subnetting is a technique for reducing IP address waste in a block. We use the host id bits of a classful IP address as the net id bits. We specify the IP address and the number of bits for the mask (usually followed by a '/' symbol), for example, 192.168.1.1/28. The subnet mask is identified by setting the specified number of bits out of 32 to 1, for example, in the given address, we need to set 28 out of 32 bits to 1 and the remainder to 0, resulting in 255.255.255.240 as the subnet mask.

Important points in Classless Addressing

- IP Addressing with Classless Addressing is a more advanced scheme.
- It improves the efficiency of IP address allocation.
- It takes the place of the older class-based addressing scheme.
- Classless Inter-Domain Routing is another name for it (CIDR).

CIDR Block

CIDR automatically assigns a block of IP Addresses depending on such rules when a user requests a certain number of IP Addresses. This block includes the number of IP addresses that the user has requested. The CIDR block is a set of IP addresses.

Rules for Creating CIDR Block-

The following three rules are used to build a CIDR block:

Rule -01:

The CIDR block's IP addresses must all be contiguous.

Rule-02:

The block's dimension must be presentable as a power of two.

The total number of IP addresses in the block is the block's size. Any CIDR block will still have a size of $2^{\text{pow}1}$, $2^{\text{pow}2}$, $2^{\text{pow}3}$, $2^{\text{pow}4}$, $2^{\text{pow}5}$, and so on.

Rule-03:

The block's first IP address must be divisible by the block's height.

CIDR Notation

CIDR IP Addresses look like-

a.b.c.d / n

- They finish with a dash and a number known as an IP network prefix.
- The IP network prefix specifies the number of bits used to identify a network.
- The remainder of the bits was used to identify hosts in the network.

Example

An example of a CIDR IP Address is-

182.0.1.2 / 28

It implies-

- The detection of the network is done with 28 bits.
- The remaining four bits are used to identify hosts in the network.

8.5 Subnetting

Subnetting is the practice of breaking up a big network into smaller networks. A single-wide network is similar to a city with no sectors or street addresses. A postman in such a town could take 3 to 4 days to locate a single address. He will conveniently locate any address in less than an hour if the town is separated into sectors and streets.

Let's look into another scenario. There will be a planned power outage due to repairs. If the town is split into sectors, the electric department will make a local announcement for the affected sector instead of a general announcement. Aside from these two examples, there are several other real-life examples of massive structures being broken into smaller parts. The same principle applies to computer networks. Subnetting is a computer networking technique that divides a wide IP network into smaller IP networks called subnets. The default class A, B, and C networks each have 16777214, 65534, and 254 hosts. When there are a large number of hosts on a single network, problems such as broadcast, collision, and congestion arise.

Let's look at a case. There are four divisions of a company: manufacturing, distribution, growth, and management. There are 50 users in each department. A private class C IP network was used by the company. Both computers can run in a single big network if there is no subnetting.



Figure 21 A single large Class C IP Network

To enter and supply information in a network, computers use transmitted signals. In a computer network, a transmitted message is an announcement message that is heard from all hosts in the network. Since all machines are connected to the same network, they can receive all broadcast signals, whether or not the broadcast messages are important to them. This network can be split into subnets in the same way as a city is divided into sectors. When a network is split into subnets, machines can only transmit broadcasts that pertain to them. Since the corporation has four departments, the network can be divided into four subnets.

Description	Network1	Network2	Network3	Network4
Network address	192.168.1.0	192.168.1.64	192.168.1.128	192.168.1.192
Valid hosts	192.168.1.1 to 192.168.1.62	192.168.1.65 to 192.168.1.126	192.168.1.129 to 192.168.1.190	192.168.1.193 to 192.168.1.254
Broadcast address	192.168.1.63	192.168.1.127	192.168.1.191	192.168.1.255

Advantages of Subnetting

- Subnetting is a technique for dividing a big network into smaller networks. Small networks are simple to administer.
- By having only the transmitted traffic that is important to the subnet, subnetting eliminates network traffic.
- Subnetting increases the network's overall efficiency by reducing unwanted traffic.
- Subnetting improves network stability by blocking a subnet's traffic inside a subnet.
- Subnetting decreases the number of IP addresses needed.

Disadvantage of Subnetting

- To connect, different subnets need an intermediary system known as a router.
- Getting more subnets means losing more IP addresses so each subnet has its network and broadcast addresses.
- Subnetting increases the network's complexity. The subnetted network must be managed by an experienced network administrator.

8.6 Network Address Translation

It's a method of converting private IP addresses to public IP addresses. It was first mentioned in RFC 1631.

What is the reason for its implementation?

NAT was implemented due to the scarcity of IP addresses. There was a time when we believed that the Internet's expansion would be halted due to the scarcity of IP addresses. If we have five networks, we want internet access to each of them. since the need for IP addresses has risen As a result, we assumed that the internet's growth would eventually slow.

The solution to the implementation of IP Addresses

However, a short-term solution to the issue of IP address depletion or exhaustion has been given. One is IP version six, which is a long-term workaround that we have already implemented. One is CIDR classless inter-domain routing, which is a short-term solution, and the other is NAT network address translation, which is a long-term solution.

NAT is a method for converting private IP addresses to public IP addresses. It's a method of preserving IP addresses. There are a variety of private addresses available as seen in Table 6.

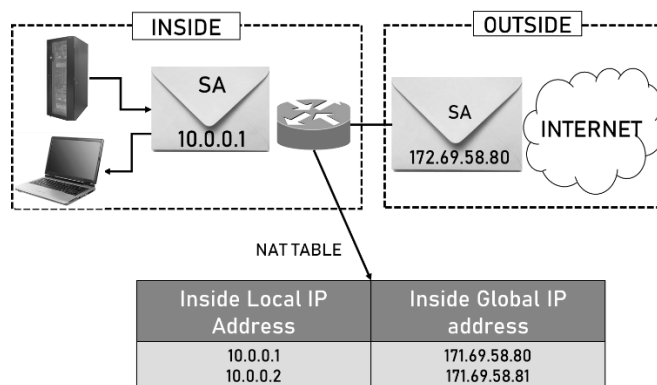
Table 6 Range of Addresses

10.0.0.0	-	255.255.255.255
172.16.0.0	-	172.32.255.255
192.168.0.0	-	192.168.255.255

A variety of private addresses are available. These IP addresses have been allocated to private networks for you to connect to the Internet.

Role of NAT

Network address conversion can translate these IP addresses into public IP addresses, allowing them to access the internet. For example, we would use NAT to allocate a single public IP address to all of the systems that have private IP addresses, allowing them to connect to the Internet.



We have a setup with one server with a private IP address of 10.0.0.1. We also have a router with NAT installed, and every router maintains a routing table with a public IP address that they use to map which private address to which public IP address, which is known as a global IP address. And that device would have a global IP address, making it visible on the internet. It will now map 10.0.0.1 to 172.69.58.80 with the aid of the router. It says that the device will not be recognized by the 10.0.0.1 server. That is a non-public IP address. 172.69.58.80 will include this information. The public IP address is that.

NAT Working

A border router has one interface in the local (inside) network and one interface in the global (outside) network and is designed for NAT. NAT transfers a local (private) IP address to a global (public) IP address as a packet travels outside the local (inside) network.

The global (public) IP address of a packet is translated to a local (private) IP address as it reaches the local network.

The packets will be lost and an Internet Control Message Protocol (ICMP) host unreachable packet to the destination will be transmitted if NAT runs out of addresses, i.e. no addresses are left in the pool setup.

What is the purpose of masking port numbers?

Assume two hosts A and B are linked in a network. Now, on the host side, they both request the same destination, on the same port number, say 1000, at the same time. If NAT just performs IP address conversion, when their packets arrive at the NAT, all of their IP addresses are disguised by the network's public IP address and sent to the destination. Destination will respond using the router's public IP address. As a result, when NAT receives a response, it would be unknown which reply belongs to which host (because source port numbers for both A and B are the same). As a result, NAT masks the source port number and creates an entry in the NAT table to prevent such a problem.

Network Address Translation (NAT) Types

There are 3 ways to configure NAT:

Static NAT – In this configuration, a single unregistered (Private) IP address is mapped to a legally registered (Public) IP address, resulting in a one-to-one mapping of local and global addresses. This is a popular term for web hosting. This is not used in businesses since many devices need Internet connectivity, and providing Internet access necessitates the use of a public IP address.

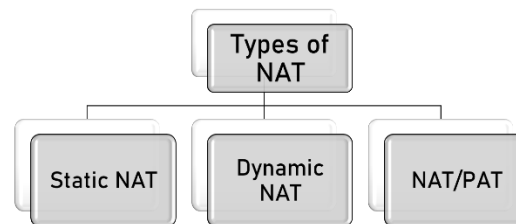


Figure 22 Types of NAT

As 3000 devices need Internet access, the organization would have to purchase 3000 public addresses, which would be very expensive.

Dynamic NAT: An unregistered IP address is converted into a registered (Public) IP address from a pool of public IP addresses in this form of NAT. The packet would be lost if the pool's IP address is not open, since only a limited number of private IP addresses can be converted to public addresses. If a pool of two public IP addresses is available, only two private IP addresses can be interpreted at any given time. Since several private IP addresses are mapped to a pool of public IP addresses, if a third private IP address tries to access the Internet, the packet will be lost. When the number of people who wish to use the Internet is set, NAT is used. This is also very expensive since the company would purchase a large number of global IP addresses to create a pool.

PAT: NAT overload is also known as Port Address Translation (PAT). Many local (private) IP addresses can be mapped to a single registered IP address using this method. The traffic is differentiated by port numbers, which identity which traffic belongs to which IP address. This is the most common method. After all, it is cost-effective because thousands of users can be connected to the Internet with only one single global (public) IP address.

Advantages of NAT

- NAT protects lawfully assigned IP addresses.
- It ensures anonymity by masking the device's IP address when sending and receiving traffic.
- When a network changes, address renumbering is no longer necessary.

Disadvantage of NAT

- Switching route delays occur as a result of translation.
- Although NAT is enabled, certain applications will not work.
- Tunneling protocols like IPsec get more complicated.
- Also, as a network layer system, the router should not tamper with port numbers (transport layer), but it does due to NAT.

ICMP (Internet control message protocol)

Which layer does ICMP Internet Control Message Protocol run in? It runs in the network layer, so which packet has the greatest chance of being discarded? What does that mean when we're sending a packet from source to destination and there's noise, or there's a congestion problem, or there's a buffer problem?

Any packets are discarded so that traffic can flow and congestion can be relieved.

Which packet has the greatest likelihood of being discarded?

The ICMP stands for Internet Managed Messaging Protocol. The first one is the ICMP then IGMP then UDP, and then TCP. Routers search for ICMP packet to discard so that they can make a space for the highest priority protocol

ICMP<IGMP<UDP<TCP

Which is the highest priority protocol?

TCP.

So, there are two situations in which we use the ICMP:

- error management and request and respond as shown in Figure 23.

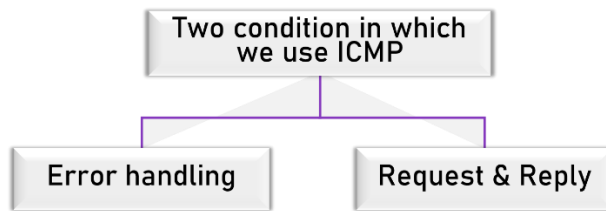


Figure 23 Two situations to use ICMP



Figure 24 ICMP(a)

If a source sends a packet, and the source is going to transmit an IP packet, the source must send the IP packet to the destination, and it must send it via Routers R1 and R2 as shown in Figure 24. So, my packet arrived at R1 first, then R2. However, when my packet reaches R2, it is rejected. My packet was suddenly discarded due to congestion or a buffer problem; there was so much traffic, and the packet was discarded. As a result, the source's IP packet has been discarded. The router, R2, is responsible for informing the source that your packet has been discarded. As a result, the packet must be resent. So, in this case, the ICMP packet (Internet Control Message Protocol) is used to provide clarification that a packet has been lost. As a result, an ICMP packet containing the source and destination addresses will be sent to the source to notify them that their packet has been discarded as seen in Figure 25. How we deal with the mistake in this situation.

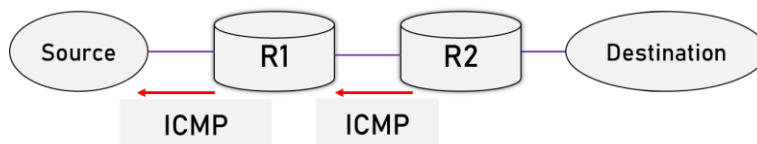


Figure 25 ICMP(b)

Request and response are the second.

Using ICMP packets, the sender will make a direct request to the destination. When studying ICMP, there are certain fundamental laws that you must adhere to. What are the basic rules? We will use the ICMP packet as input if the IP packet is discarded.

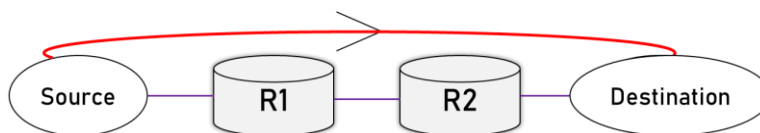


Figure 26 ICMP Request and Reply

Some Basic Rules

If IP packet is discarded

- We can use the ICMP packet as feedback.

If the ICMP packet is discarded

- No ICMP packet will be used as feedback.

However, let's use the ICMP packet as an example to see what the issue is. So, what sort of issue might it be? We've got a starting point and an endpoint. The IP packet will be sent to routers r1 and r2, and it is assumed that R2 will discard the packet due to a congestion problem. Now R2 is going to use the feedback, and which feedback packet it uses, it uses the ICMP packet as feedback. (R2, S) is written in brackets in this case. S is the destination to whom it wishes to warn that your packet has been lost, and R2 is the source one and second. When it arrived at r1, however, r1 still discarded the packet due to a congestion problem or buffer. R1 now has to notify R2 that the packet you sent for the source has been discarded. In this case, the source is R1 and the destination is R2. As a result of the ICMP comments, the packet will be sent to R2 once more. Still, owing to some congestion, R2 has a lot of congestion and a lot of other protocols, and R2's highest protocols are TCP and if ICMP packets are present, it will discard the lowest priority one, which is the ICMP packet.

So R2 will discard the ICMP packet; in this situation, R2 must remind r1 that whatever packet you sent me, whatever information it contained, has been lost. But, in this case, R2 sends the information to r1, where r2 is the source and r1 is the destination. It's sending to r1, so r1 can discard the packet once more. Since ICMP is the packet with the lowest priority. If there is congestion, they can dump the ICMP envelope, which has the lowest priority. Now, using an ICMP packet, r1 can send a feedback type or feedback request to R2. R2 will discard once more, and this will continue to happen over and over. So, endless loop, what do you think is going on here? As a result of the law violation, you will be trapped in an endless loop as shown in Figure 27.

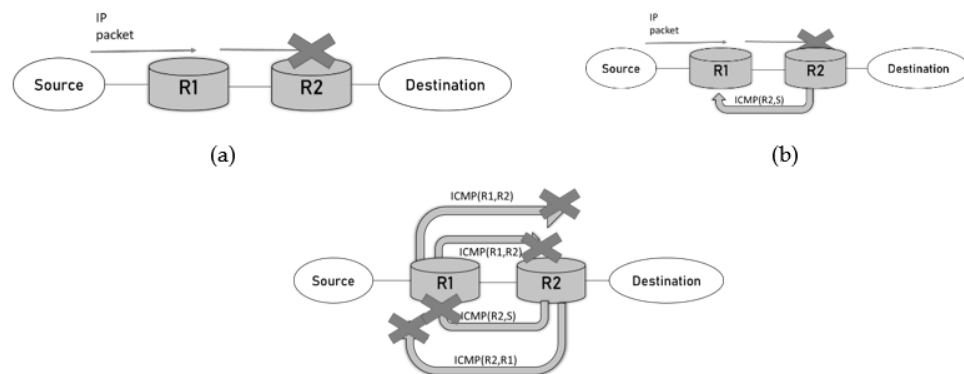


Figure 27 Issue in using ICMP packet

Important point

Always note that if you are sending an ICMP packet and it is misplaced, you cannot use an ICMP packet as feedback; you must use another packet. If an IP packet is refused, it is thrown out. The ICMP packet can be used as input. However, if an ICMP packet is destroyed, it cannot be used as feedback.

8.7 Address Resolution protocol

To send/receive messages, most computer programs/applications use a logical address (IP address), but the direct correspondence takes place over a physical address (MAC address), which is layer 2 of the OSI model. As a result, our goal is to obtain the destination MAC address, which is needed for communication with other devices. This is where ARP comes into play; it aims to convert IP addresses to physical addresses as seen in Figure 28.

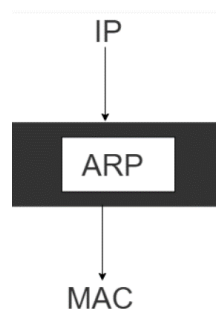


Figure 28 Address Resolution Protocol

ARP stands for Address Resolution Protocol and is one of the most relevant protocols in the OSI model's Network layer. From a host's established IP address, ARP determines the hardware address, also known as the Media Access Control (MAC) address.

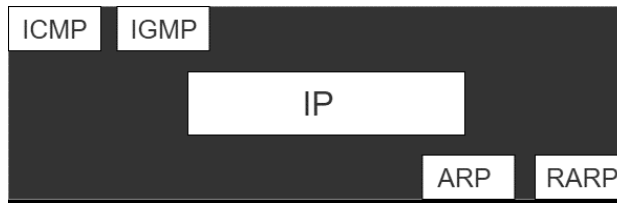


Figure 29 Network Layer

ARP Working

Consider a computer that wishes to connect with another device over the internet. What is the purpose of ARP? Is it broadcasting a packet to all of the originating network's devices? The network devices peel the data link layer header from the protocol data unit (PDU) called frame and send the packet to the network layer (OSI layer 3), where the packet's network ID is validated against the destination IP's network ID, and if they are identical, it responds to the source with the destination's MAC address. The above procedure is repeated until the second last network computer in the path to the destination is validated, at which point ARP responds with the destination MAC address.

8.8 Reverse Address Resolution Protocol

Reverse Address Resolution Protocol (RARP) is an acronym for Reverse Address Resolution Protocol, which is a computer networking protocol that allows a client computer to obtain its IP address from a gateway server's Address Resolution Protocol table or cache. In the gateway-router, the network administrator builds a table that maps the MAC address to the corresponding IP address. This protocol is used to transfer data between two server points. The client does not need to know who the server is capable of serving its request in advance. An administrator must configure each server's Medium Access Control (MAC) addresses individually. RARP is only capable of supporting IP addresses. When a replacement computer is set up, it can or may not have a connected disc that can hold the IP Address indefinitely, so the RARP client software must request the IP Address from the router's RARP server. Under the assumption that an entry has been created in the router table, the RARP server will return the IP address to the machine.

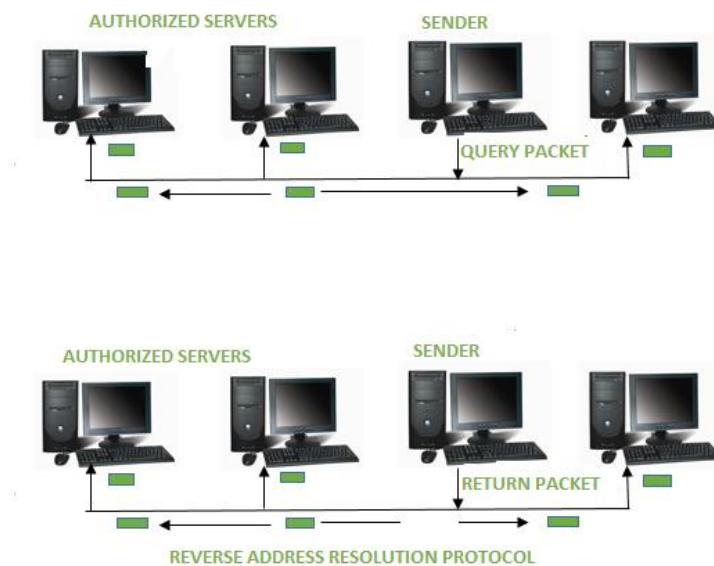


Figure 30 Reverse Address Resolution Protocol

RARP Working

The RARP is a Network Access Layer protocol that allows data to be sent between two points in a network.

Each network member has two distinct addresses: an IP (logical) address and a MAC (physical) address (the physical address). The IP address is allocated by program, and the MAC address is built into the hardware after that.

Any ordinary device on the network will act as a RARP server and respond to RARP requests. It must, however, store the data of all MAC addresses as well as their allocated IP addresses. If the network receives a RARP order, only these RARP servers can respond. The data packet must be transmitted over very low-cost network layers. This means that the package is sent to each of the participants at the same time. For an Ethernet broadcast address and its physical address, the client sends out a RARP order. The server then informs the client of its IP address.

Summary

- A global identification scheme that uniquely identifies every host and router is required at the network layer for packet transmission from host to host.
- An IPv4 address is 32 bits long and identifies a host or router on the Internet specially and universally.
- The netid is the part of the IP address that defines the network in classful addressing.
- The part of the IP address that specifies the host or router on the network is known as the host in classful addressing.
- The relation of a computer to a network is defined by its IP address.
- IPv4 addresses are divided into five categories. The number of hosts per network permitted in Classes A, B, and C varies. Multicasting is in Class D, and Class E is reserved.
- The first byte of an address can conveniently be used to evaluate its class.
- For unicast correspondence, addresses in classes A, B, or C are often used.
- Multicast correspondence is carried out using addresses in class D.
- Subnetting splits a wide network into many smaller ones, introducing an intermediate level of IP addressing hierarchy.
- We can partition the address space into variable-length blocks with classless addressing.
- IPv6 addresses are divided into three categories: unicast, anycast, and multicast.
- Reverse Address Resolution Protocol (RARP) is an acronym for Reverse Address Resolution Protocol, which is a computer networking protocol that allows a client computer to obtain its IP address from a gateway server's Address Resolution Protocol table or cache.

Keywords

IP Address: An Internet Protocol address (IP address) is a numerical mark assigned to each interface (e.g., computer, printer) in a computer network that communicates using the Internet Protocol.

IP Protocol: The Internet Protocol (IP) is the principal communications protocol used for relaying datagrams (packets) across an internetwork using the Internet Protocol Suite.

NAT: Network address translation (NAT) is a way of converting one IP address space to another by changing network address information in packets' IP headers as they are in transit via a traffic routing system.

ARP: ARP stands for Address Resolution Protocol, and is one of the most relevant protocols in the OSI model's Network layer.

RARP: Where only the ethernet address is identified and the IP address is needed, RARP offers the opposite service to ARP.

PAT: Different devices on a LAN can be mapped to a single public IP address using Port Address Translation (PAT), which is an extension of Network Address Translation (NAT).

Classful Addressing: The address space is divided into five classes of Classful addressing: A, B, C, D, and E. Each of these classes has an IP address set that is valid.

Classless Addressing: Unlike classful addressing, classless IPv4 addressing does not partition the address space into classes.

Subnetting: Subnetting is the process of dividing a larger network into smaller networks to preserve security.

Review Questions

1. Explain the IP Protocol. What makes it different from the TCP protocol?
2. What are IP addresses, and what do they mean? Describe how an IP address is formatted.
3. Distinguish between IPV4 and IPV6 addressing, as well as their grouping.
4. Explain what subnetting is and how it works.
5. Difference between ARP and RARP.

Self Assessment

1. What is the format of the IP address?
 - a) 34 bit
 - b) 64 bit
 - c) 16 bit
 - d) 32 bit
2. Version 6 of the IP address has how many bits.
 - a) 64 bits
 - b) 128 bits
 - c) 32 bits
 - d) 256 bits
3. How many versions/s of IP's are there?
 - a) 4 versions
 - b) 3 versions
 - c) 2 versions
 - d) 1 version
4. Which technology allows a large number of private IP addresses to be represented by a smaller number of public IP addresses?
 - a) NAT
 - b) NTP
 - c) RFC 1631
 - d) RFC 1918
5. What is the effect of the overload keyword in a static NAT translation configuration?
 - a) It enables port address translation.
 - b) It enables the use of a secondary pool of IP addresses when the first pool is depleted.
 - c) It enables the inside interface to receive traffic.
 - d) It enables the outside interface to forward traffic.
6. What is the first step in the NAT configuration process?

- a) Define inside and outside interfaces.
 - b) Define public and private IP addresses.
 - c) Define IP address pools.
 - d) Define global and local interfaces.
7. Which of the following is the ethernet broadcast address used in ARP and RARP requests?
- a) 255.255.255.255
 - b) 08:00:20:11:aa:01
 - c) ff:ff:ff:ff:ff:ff
 - d) 224.0.0.0
8. Which of the following describes the function of ARP?
- a) It is used to map a 32-bit IP address to a 48-bit ethernet address.
 - b) It is used to map a 48-bit ethernet address to a 32-bit IP address.
 - c) It is used to map a 32-bit ethernet address to a 48-bit IP address.
 - d) It is used to map a 48-bit IP address to a 32-bit ethernet address.
9. In classful addressing, the address space is divided into
- a. Five classes
 - b. Ten classes
 - c. Fifteen classes
 - d. Three classes
10. In classless addressing, there are no classes but the addresses are still granted in
- a. Sections
 - b. Blocks
 - c. Codes
 - d. All of the above.
11. In IPv4 Addresses, classful addressing is replaced with
- a) Classless Addressing
 - b) Classful Addressing new version
 - c) Classful Advertising
 - d) Classless Advertising
12. The first address in a block is used as the network address that represents the
- a) Class Network
 - b) Entity
 - c) Organization
 - d) DataCodes

Answers

- | | | | | | | | |
|----|---|-----|---|-----|---|-----|---|
| 1. | a | 2. | b | 3. | c | 4. | a |
| 5. | a | 6. | a | 7. | c | 8. | a |
| 9. | a | 10. | b | 11. | a | 12. | c |

Further Readings



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